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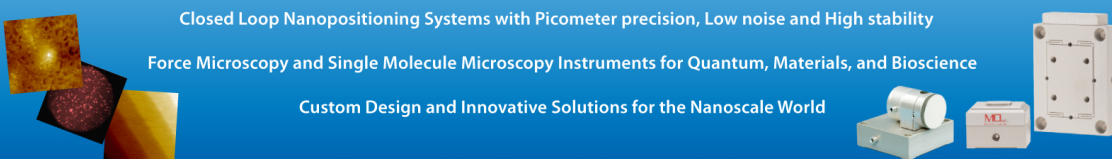
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Energy loss of 1–50 keV H, He, C, N, O, Ne, and Ar ions transmitted through thin carbon foils

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Thin foils are used extensively in space plasma spectrometers, for example, for generation of a coincidence measurement via secondary electron emission in time-of-flight mass spectrometers and charge conversion of energetic neutral atoms (ENAs) in ENA imagers. An ion or neutral atom passing through the carbon foil experiences energy loss and straggling that can degrade the energy resolution or mass resolution. These effects need to be quantified in order to permit efficient instrument development and modeling. We present measurements of energy loss and energy loss straggling for H, He, C, N, O, Ne, and Ar ions between 1 and 50 keV passing through carbon foils of different thicknesses (nominally between 0.5 and 3.0 $\mu\text{g}/\text{cm}^2$). We derive empirical relationships for the energy loss and energy loss straggling as a function of foil thickness, ion species, and ion energy. In particular, below ~ 10 keV the energy loss of hydrogen seems to vary with foil thickness and the energy loss straggling seems to vary with the square root of the thickness. Our results apply equally well to ions and neutral atoms. © 2006 American Institute of Physics.

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I. INTRODUCTION

Thin carbon foils are critical components of many space instruments used for time-of-flight (TOF) mass spectrometry^{1,2} and energetic neutral atom (ENA) imaging.^{3,4} In TOF mass spectrometry, ions transit a foil at the entrance of the drift region, where they generate secondary electrons that are used to generate a start pulse that starts the TOF timer. At the other end of the drift region, the ion itself is detected, stopping the timer and providing the TOF, and therefore velocity, of the ion. In some types of neutral atom imagers, the foil converts ENAs to ions, thus allowing them to be electro-optically filtered and ultimately detected. Besides these useful properties of electron emission and charge exchange, however, the interaction also has unwanted side effects such as energy loss and straggling as well as angular scattering, all of which are strong functions of particle species and energy as well as foil composition, thickness, and impact angle. These effects can impact the performance of the instrument, specifically the mass resolution, energy resolution, and sensitivity. Because these undesirable factors scale with the thickness of the foils, a significant effort has been made to develop a means for handling and supporting extremely thin (<10 nm) foils.²

In the perspective of space instrument development and modeling, a good characterization of the interaction between the incident ions or ENAs and the carbon foils is critical for understanding instrument performance. So far, the technical literature in the keV range, e.g., Refs. 5–10, is often not consistent because there is no systematical study on both

energy loss and energy straggling for a broad range of ion energies and species and foil thicknesses. Therefore, empirical scaling laws for different foil thicknesses are difficult to determine from existing data. There is a need for a consistent set of data taken with the same instrumentation for a set of carbon foils and a collection of projectile species.

The goal of this article is to provide a useful and consistent set of data from which systematic understanding and quantitative relationships between the ion beam parameters and foil thickness can be developed. These data and empirical laws derived from them should provide a useful tool for accurate modeling and development of space plasma instrumentation.

The experimental setup is described in the next section. We then present measurements of energy loss and straggling of H, He, C, N, O, Ne, and Ar ions passing through thin (nominal 0.5–3.0 $\mu\text{g}/\text{cm}^2$) carbon foils in the following section.

II. EXPERIMENTAL SETUP

The ion beam is produced by a duo-plasmatron ion source. A Wien filter selects the ion mass per charge with a mass resolution of $M/\Delta M \sim 50$. The beam energy ranges from ~ 500 eV/ Q up to 51 keV/ Q . The acceleration potential is known to about ± 0.01 kV above 20 kV and ± 0.005 kV below 20 kV. An upper limit of 15 eV for the energy width of the beam was determined with an energy analyzer with a 0.5% passband. The measurements were performed in an ultraclean vacuum chamber at pressures below 10^{-7} Torr.

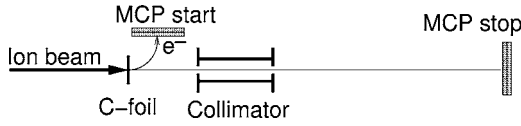


FIG. 1. Schematic of the experimental setup. The collimator limits the field of view to $\pm 2.7^\circ$.

The energy loss measurements were made with a specially constructed time-of-flight system. More details can be found in a forthcoming publication.¹¹ Figure 1 shows a schematic view of the experimental setup. The ion beam is directed onto a carbon foil. Our carbon foils were purchased from the Arizona Foil Company. We used different nominal (as cited by the manufacturer) thicknesses ranging from 0.5 up to $3.0 \mu\text{g}/\text{cm}^2$. The foils were mounted on 333 lpi (lines per inch) nickel grids (70% transmission) manufactured by Buckbee-Mears.

The secondary electrons emitted from the exit surface, resulting from the interaction of the incident ions with the carbon foil,^{12,13} are electrostatically steered to the start microchannel plate (MCP) assembly. The detection of the electron(s) starts a timer for the TOF measurement. The potential difference between the carbon foil (at ground potential) and the front plate of the start MCP assembly is defined as V_{bias} . An electrostatic simulation performed with SIMION (Ref. 14) showed that the deflection of the beam after the carbon foil due to V_{bias} is negligible for a beam energy per charge above a few keV/q . The ions or neutrals pass through a collimator made of two pinholes that set a maximum beam divergence of $\pm 2.7^\circ$. The collimator is used to limit the measurements to small angle scattering. The flight path length is $208.65 \pm 0.30 \text{ cm}$. The ions or neutrals are detected by the stop MCP assembly and stop the timer. The time difference between the stop and the start signals is the TOF, which is converted to energy because the ion beam mass is known.

The MCP assemblies are chevron stacks of two individual plates. MCP signals are extracted by capacitive decoupling and amplified outside the vacuum chamber. TOF spectra were acquired with a Fast ComTec P7888 card with 1 ns resolution. The card was calibrated with a delay time pulse generator. Start and stop signal cables are identical and have the same length to ensure that the time delay of the start and stop signals are identical and have the same length.

A. Contribution of secondary electron time dispersion to ion energy measurements

In our experiment the time of flight (τ_{electron}) of secondary electrons from their emission at the back surface of the foil to their detection at the start detector is a non-negligible contribution to the measurement of the TOF of the ion. The true TOF (τ) of the primary ion in the drift tube is therefore $\tau = \tau_{\text{meas}} + \tau_{\text{electron}}$. The secondary electron TOF can be inferred by using two properties: (a) the average electron TOF only depends on the acceleration voltage bias (V_{bias}) applied to the front plate of the start MCP assembly, and (b) the average energy loss of an ion that transits the foil is independent of V_{bias} . Constraint (a) is reasonable because, in the case of metals, for example, the secondary electron energy spec-

trum is independent of the primary particle energy for protons,¹⁵ so the transport of electrons to the start detector depends only on the electrostatic acceleration of the electron to the detector. Thus, by comparing ion TOF measurements performed at different energies and at different V_{bias} we can accurately determine the electron TOF. The energy loss ΔE is given by

$$\Delta E = E_0 - E_1 = E_0 - \frac{ms^2}{2(\tau_{\text{meas}} + \tau_{\text{electron}})^2}, \quad (1)$$

where E_0 and E_1 are the incident and residual energies respectively, m is the mass of the ion or ENA, and s is the TOF path length.

The method is based on a least square minimization of the variance of ΔE for different incident energies and V_{bias} . The parameters of the minimization are the respective τ_{electron} . The result of this method provides the best electron TOF for each V_{bias} and the energy loss for each energy. Strictly speaking, any systematic contribution to the TOF measurement, such as a systematic delay introduced by the electronics or cable length difference, would be included in the term τ_{electron} by using the method we just described. In order to give an idea of the accuracy we obtained with this method, we find a mean value of $\langle \tau_{\text{electron}} \rangle = 8.94 \pm 0.02 \text{ ns}$ for $V_{\text{bias}} = 600 \text{ V}$.

The time dispersion of τ_{electron} is another effect that we need to take into account when determining energy straggling. Primary energy straggling results in a broadening of the TOF peak. In the case of Gaussian-like distributions the contributions simply add quadratically. In our case most of the TOF distributions are asymmetric (or skewed) and deviate from a Gaussian character. Figure 2(a) shows an example of a TOF spectrum for 4 keV protons passing through a nominal $0.5 \mu\text{g}/\text{cm}^2$ carbon foil before (dashed) and after (solid) correction for electron TOF. We assume, however, that the contributions to TOF dispersion add quadratically. Thus, the measured TOF full width at half maximum (FWHM) is the quadratic sum (neglecting cross terms) of the true TOF FWHM (due to energy straggling) and the secondary electron TOF dispersion:

$$(\Delta \tau_{\text{meas}})^2 = (\Delta \tau)^2 + (\Delta \tau_{\text{electron}})^2. \quad (2)$$

Because $\Delta \tau_{\text{electron}}$ mainly depends on the acceleration voltage V_{bias} (e.g., $\tau_{\text{electron}} \rightarrow 0$ and $\Delta \tau_{\text{electron}} \rightarrow 0$ as $V_{\text{bias}} \rightarrow \infty$), we experimentally determined $\Delta \tau_{\text{electron}}$ by varying V_{bias} . From Eq. (2) we can write

$$\begin{aligned} (\Delta \tau)^2 &= [\Delta \tau_{\text{meas}}(V_1)]^2 - [\Delta \tau_{\text{electron}}(V_1)]^2 \\ &= [\Delta \tau_{\text{meas}}(V_2)]^2 - [\Delta \tau_{\text{electron}}(V_2)]^2, \end{aligned} \quad (3)$$

where V_1 and V_2 are two different values of V_{bias} . Equation (3) can be rewritten as

$$\begin{aligned} [\Delta \tau_{\text{meas}}(V_1)]^2 - [\Delta \tau_{\text{meas}}(V_2)]^2 \\ = [\Delta \tau_{\text{electron}}(V_1)]^2 - [\Delta \tau_{\text{electron}}(V_2)]^2 \end{aligned} \quad (4)$$

and, hence,

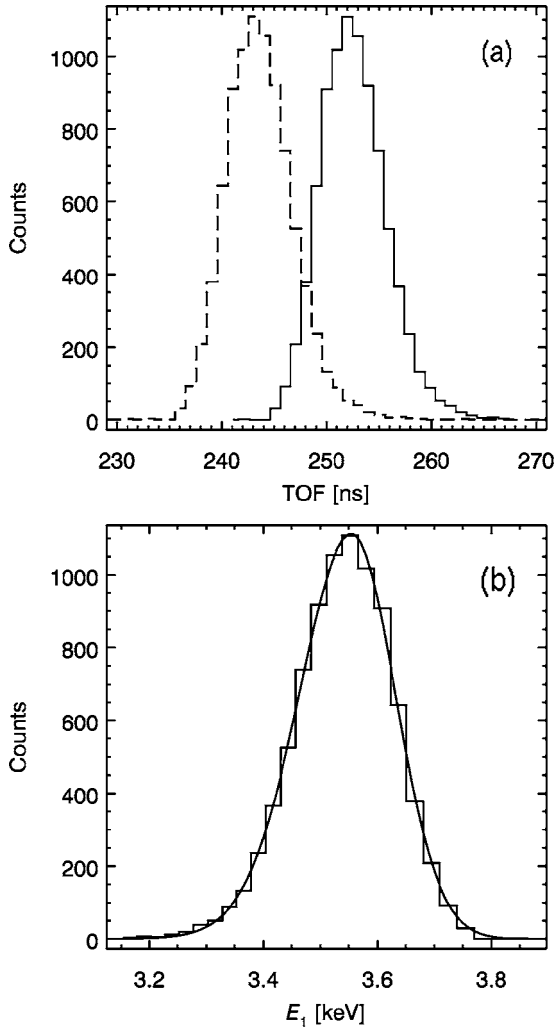


FIG. 2. (a) Typical measured (dashed) and corrected (solid) TOF spectrum for 4 keV protons passing through a (nominal) $0.5 \mu\text{g}/\text{cm}^2$ carbon foil. One TOF channel corresponds to 1 ns. (b) Corresponding energy spectrum after correcting for all experimental effects.

$$\begin{aligned} & [\Delta\tau_{\text{meas}}(V_1)]^2 - [\Delta\tau_{\text{meas}}(\infty)]^2 \\ &= [\Delta\tau_{\text{electron}}(V_1)]^2 - \underbrace{[\Delta\tau_{\text{electron}}(\infty)]^2}_{\rightarrow 0} = [\Delta\tau_{\text{electron}}(V_1)]^2. \end{aligned} \quad (5)$$

We plot the square root of the left hand side quantity of Eq. (4) with $V_1=100$ V as a function of V_{bias} in Fig. 3. The quantity $\sqrt{[\Delta\tau_{\text{meas}}(V_1)]^2 - [\Delta\tau_{\text{meas}}(\infty)]^2}$ is the asymptotic value a_0 of the curve, which is a fit using the following function:

$$F(V_{\text{bias}}) = a_0(1 - \{1/\exp[a_1(x - a_2)]\}).$$

Using Eqs. (3) and (5) and replacing the values, we can readily calculate the secondary electron TOF dispersion for the voltage V_2 . For a voltage bias of 600 V we find $\Delta\tau_{\text{electron}}(V_{\text{bias}}=600 \text{ V})=1.14 \text{ ns}$.

After taking into account all the experimental effects, we can derive a “true” TOF spectrum [histogram with solid line of Fig. 2(a)] that can be readily transformed into an energy spectrum using the relation $E_1=(\frac{1}{2})m(s/\tau)^2$, where E is the incident ion energy, m is its mass, s is the TOF distance, and

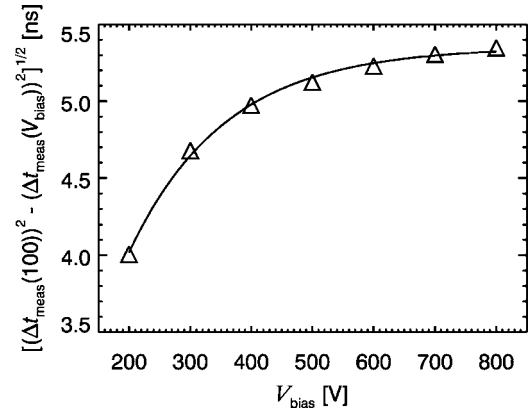


FIG. 3. Illustration of the method to infer the electron TOF dispersion. See text for explanation.

τ is the true TOF of the particle. Figure 2(b) shows the corresponding energy spectrum. The curve is a fit $C(E) = C_0 \cdot f(E)$, where $f(E)$ is a kappa distribution for $E \leq \hat{E}$ and a Gaussian distribution for $E > \hat{E}$ as follows:

$$f(E) = \begin{cases} \left[1 - \frac{(E - \hat{E})^2}{2\sigma_K^2 b} \right]^{-b}, & E \leq \hat{E} \\ \exp\left[-\frac{(E - \hat{E})^2}{2\sigma_G^2} \right], & E > \hat{E}. \end{cases} \quad (6)$$

This function fits most of our spectra very well. Nevertheless, since it requires five free parameters, fits of narrower peaks do not work as well. If we apply the fit to the residual energy spectrum, the parameter $\hat{E}$ represents the most probable residual energy. For the special case where $b=2$ the kappa distribution becomes a Lorentzian distribution, and if $b \rightarrow \infty$, the distribution becomes Gaussian. The parameter σ_G is the sigma of the Gaussian distribution.

III. MEASUREMENTS OF FOIL THICKNESS

It is well known, e.g., Refs. 4 and 16, that the actual thickness of ultrathin carbon foils is larger than the nominal thickness. Foil thickness can be estimated by measuring the angular scattering half-width at half maximum $\psi_{1/2}$, of the scatter distribution of incident ions as a function of energy. A semiempirical theory for multiple, small-angle scattering¹⁷ gives the following relation: the product of the ion energy and angular scattering width is a constant for a particular ion species, foil composition, and thickness and assuming that energy loss in the foil is much less than the incident ion energy. Therefore, we derive the foil constant $k_F = \psi_{1/2} E_0$, where E_0 is the energy of an incident beam of protons, as a measure of foil thickness.⁴ We inferred the measured foil thickness based on $1.1 \mu\text{g}/\text{cm}^2 = 12.6 \text{ keV deg}$,⁴ i.e., $t_m(\mu\text{g}/\text{cm}^2) = 0.08 k_F (\text{keV deg})$. Using this technique, Fig. 4 shows the measured thicknesses as a function of the nominal value given by the manufacturer. We notice a y-intercept value of about 1.6 ($k_F = \sim 18 \text{ keV deg}$) as if the foil thickness was systematically increased with a residue layer. In fact, foils can retain a significant amount of the parting agent used

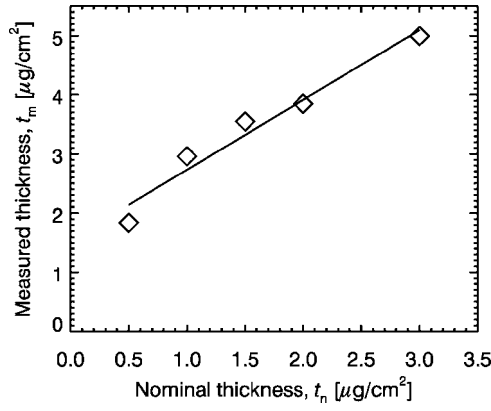


FIG. 4. Data points show the measured thickness t_m as a function of the nominal value t_n given by the manufacturer. The line is a least-square linear fit to the data points. The nonzero y-intercept value indicates the presence of adsorbates and/or parting agent residue in addition to the carbon foil.

in the manufacturing process. Moreover, because carbon is a very effective adsorbent, the surfaces might also be covered with adsorbates.

IV. MEASUREMENTS OF ION ENERGY LOSS

With this understanding, we then performed energy loss measurements for H, He, C, N, O, Ne, and Ar between 1 and 50 keV transmitted through carbon foils using different foil thicknesses.

The well-known result of the Lindhard-Scharff-Schiøtt¹⁸ (LSS) theory is that the electronic stopping power is proportional to velocity, i.e., $(dE/dx)_{\text{electronic}} \propto kv$ for $v < v_0 Z^{2/3}$, where v_0 is the Bohr velocity ($v_0 \cong 2.2 \times 10^8$ cm/s) and Z is the atomic number of the projectile. Here, k is a constant that depends on the ion species and foil material. Since our measurements were limited to low angle ($\pm 2.7^\circ$) scattering events for which the energy lost to Rutherford-type collisions is minimal, we can assume that we mainly measure the energy loss due to electronic stopping processes in which energy is lost by interactions with target electrons. For simplicity, we also assume that the stopping power is roughly equal to the average energy loss divided by the foil thickness, which is a good approximation if the energy loss ΔE of an ion in the foil is only a small fraction of the ion's incident energy. With the two assumptions above, one can relate the average energy loss to the electronic stopping power as follows:

$$\frac{dE}{dx} \approx \frac{E_0 - E_1}{t_m} \propto kv \propto k \sqrt{\frac{E}{m}} \approx k \sqrt{\frac{E_0 + E_1}{2m}}, \quad (7)$$

where E_0 and E_1 are the energies before and after the carbon foil, respectively, t_m is the measured foil thickness, and m is the ion mass. Thus, we define

$$k = \frac{E_0 - E_1}{t_m} \left(\frac{E_0 + E_1}{2m} \right)^{-1/2}. \quad (8)$$

Figure 5 shows k as a function of the mean energy of the ions.

With the proportionality of the LSS theory, one expects k to be constant. However, deviation from a flat line at low

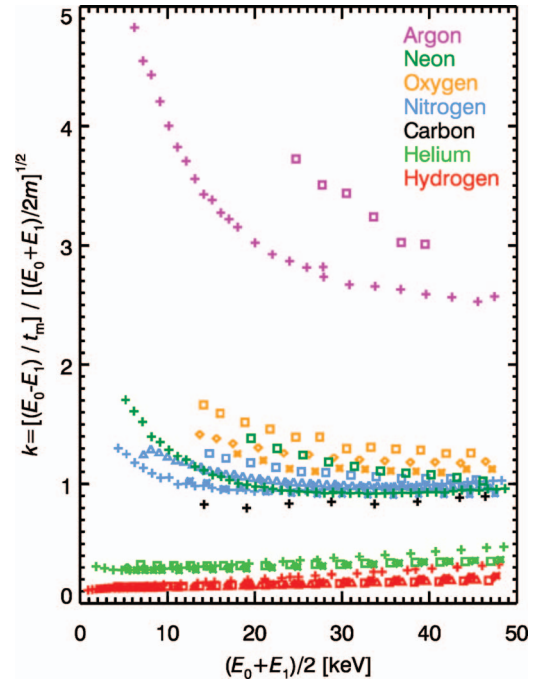


FIG. 5. (Color) k is an approximation of the electronic stopping power divided by the mean velocity. k is plotted as a function of the mean energy. Measured (t_m) and nominal (t_n) thicknesses are +, $t_m=1.84$, $t_n=0.5$; *, $t_m=2.96$, $t_n=1.0$; $\diamond$, $t_m=3.55$, $t_n=1.5$; $\triangle$, $t_m=3.85$, $t_n=2.0$; and $\square$, $t_m=5.00$, $t_n=3.0$ $\mu\text{g}/\text{cm}^2$.

velocities is a sign that the nuclear stopping is becoming a significant component, in particular in the case of argon and neon. For H and He, the increasing values of k indicate a deviation from the LSS velocity proportional electronic stopping power. The maximum stopping power for H in carbon is reached around ~ 80 keV according to SRIM-2003,¹⁹ i.e., dE/dx is constant around ~ 80 keV. Therefore, k should decrease with increasing energy already at the end of our energy range.

Oxygen has a higher energy loss than ions of adjacent atomic numbers (N or Ne). This is in agreement with the so-called Z_1 oscillation discovered by Ormrod *et al.*²⁰ The stopping power depends on the detailed shell structure and the resulting effective charge of the moving ions.

In the upper panel of Fig. 6 we show the average energy loss as a function of the initial energy for protons passing through carbon foils of five different thicknesses. As expected, the average energy loss increases with foil thickness. The lower panel of Fig. 6 shows the same measurements divided by the foil thickness to the power α , i.e., $\Delta E/t_m^\alpha$ as a function of E_0 , where $\alpha=0.77$ for hydrogen. Except for the thinnest foil ($1.84 \mu\text{g}/\text{cm}^2$, nominally $0.5 \mu\text{g}/\text{cm}^2$), the energy loss varies linearly to a reasonable extent with foil thickness to the power α . The quantity $\Delta E/t_m^\alpha$ for the thinnest foil shows a rapid departure from proportionality. A possible explanation could be the layers of impurities sticking on the surfaces of the foil, which would have a proportionately larger effect at lower nominal thicknesses. These layers might also have a higher stopping power than carbon. As a consequence, the thinnest foil would have a higher stopping power, which is what we observe. The black curves are polynomial fits to the data according to the following relation:

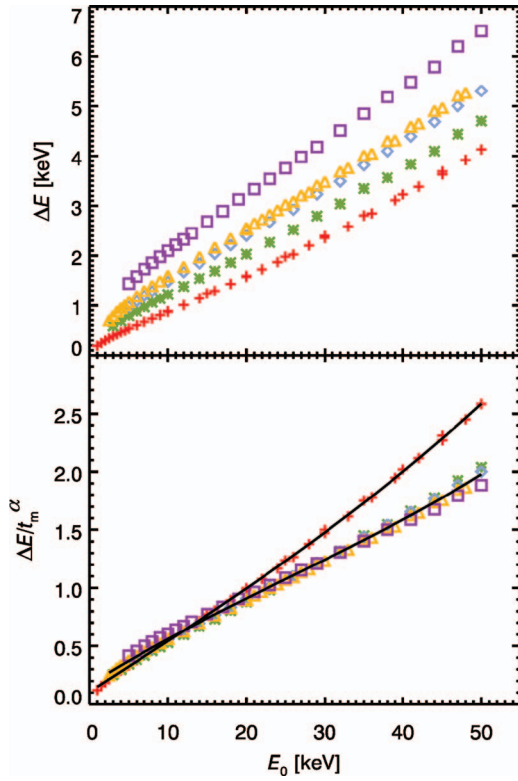


FIG. 6. (Color) Average energy loss (upper panel) and average energy loss divided by the foil thickness to the power $\alpha=0.77$ (lower panel) for hydrogen through carbon foils of different thicknesses. Measured (t_m) and nominal (t_n) thicknesses are +, $t_m=1.84$, $t_n=0.5$; *, $t_m=2.96$, $t_n=1.0$; $\diamond$, $t_m=3.55$, $t_n=1.5$; $\triangle$, $t_m=3.85$, $t_n=2.0$; and $\square$, $t_m=5.00$, $t_n=3.0 \mu\text{g}/\text{cm}^2$.

$$\Delta E/t_m^\alpha = a_0 + a_1 E_0 + a_2 E_0^2 + a_3 E_0^3. \quad (9)$$

The exponent α and the coefficients a_0 , a_1 , a_2 , and a_3 for hydrogen and nitrogen are given in Table I.

Figure 7 (upper panel) shows the energy straggling ΔE_{FWHM} (defined here as the full width at half maximum of the energy loss distribution) divided by the initial energy E_0 as a function of the initial energy. Figure 7 (lower panel) shows the same, but divided by $\sqrt{t_m}$. The energy straggling seems to be independent of foil thickness above ~ 20 – 25 keV (Fig. 7, upper panel). On the contrary, below ~ 10 keV it appears that the energy straggling varies as $\sim \sqrt{t_m}$ (Fig. 7, lower panel). There is clearly a transition region where the quantity $\Delta E_{\text{FWHM}}/E_0$ is minimum. Below and above the transition region the quantity $\Delta E_{\text{FWHM}}/E_0$ follows power laws.

It is interesting to note that the energy range (up to ~ 10 keV) where the energy loss seems to vary linearly with

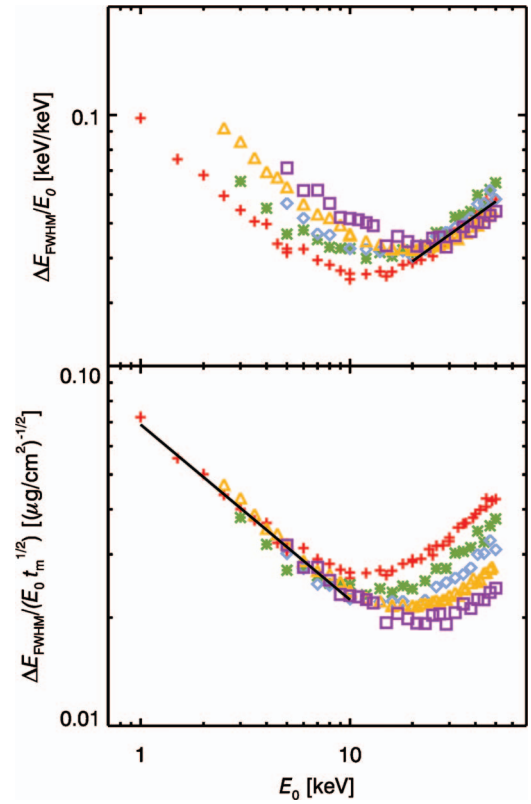


FIG. 7. (Color) Energy straggling ΔE_{FWHM} divided by the initial energy E_0 as a function of the initial energy (upper panel). In the lower panel we show the energy straggling divided by the initial energy normalized to the square root of the foil thickness. Measured (t_m) and nominal (t_n) thicknesses are +, $t_m=1.84$, $t_n=0.5$; *, $t_m=2.96$, $t_n=1.0$; $\diamond$, $t_m=3.55$, $t_n=1.5$; $\triangle$, $t_m=3.85$, $t_n=2.0$; and $\square$, $t_m=5.00$, $t_n=3.0 \mu\text{g}/\text{cm}^2$.

the thickness is also where the energy straggling varies as $\sim \sqrt{t_m}$ (Fig. 7, lower panel). The black curves are fits to the data with the following relations:

$$\Delta E_{\text{FWHM}}/E_0 = a_0 E_0^{a_1} \quad \text{for } \sim 20 < E_0 < \sim 50 \text{ keV}, \quad (10)$$

$$\Delta E_{\text{FWHM}}/(E_0 \sqrt{t_m}) = b_0 E_0^{b_1} \quad \text{for } \sim 1 < E_0 < \sim 10 \text{ keV}. \quad (11)$$

The coefficients a_0 , a_1 , b_0 , and b_1 are given in Table II. The fit (not shown here) to the nitrogen straggling data between ~ 5 and ~ 50 keV is done with Eq. (11) and the coefficients are also given in the table.

One may ask if the energy loss and straggling change with the angle of emergence of the primaries. Lennard and Geissel²¹ measured the energy loss and energy loss strag-

TABLE I. Coefficients for the fit to the energy loss.

Element	t_m ($\mu\text{g}/\text{cm}^2$)	α	a_0	a_1	a_2	a_3
H	1.84	0.77	$(1.07 \pm 0.06) \times 10^{-1}$	$(4.35 \pm 0.13) \times 10^{-2}$	$(0.1 \pm 6.1) \times 10^{-6}$	$(2.4 \pm 0.8) \times 10^{-6}$
	2.96–5.00	0.77	$(1.68 \pm 0.13) \times 10^{-1}$	$(4.25 \pm 0.22) \times 10^{-2}$	$(-3.7 \pm 1.0) \times 10^{-4}$	$(4.9 \pm 1.3) \times 10^{-6}$
N	1.84	1.27	$(5.33 \pm 0.06) \times 10^{-1}$	$(1.84 \pm 0.08) \times 10^{-2}$	$(1.06 \pm 0.34) \times 10^{-4}$	$(-7.8 \pm 4.1) \times 10^{-7}$
	2.96–5.00	1.27	$(4.76 \pm 0.24) \times 10^{-1}$	$(2.28 \pm 0.28) \times 10^{-2}$	$(-2.87 \pm 0.97) \times 10^{-4}$	$(2.7 \pm 1.1) \times 10^{-6}$

TABLE II. Coefficients for the fit to the straggling.

Element	E_0 range (keV)	Eq.	a_0 or b_0	a_1 or b_1
H	$\sim 1-10$	(11)	$(6.9 \pm 0.2) \times 10^{-2}$	-0.49 ± 0.02
	$\sim 20-50$	(10)	$(1.1 \pm 0.1) \times 10^{-2}$	0.42 ± 0.03
N	$\sim 5-50$	(11)	$(5.1 \pm 0.4) \times 10^{-1}$	-0.83 ± 0.03

gling for heavy ions through carbon foils. They find that at $v=0.8v_0$ (~ 16 keV/amu) the energy loss straggling of Mg is roughly constant up to $\sim 6^\circ$ emergence angle and increases for larger angles. In our experiment, we measure the particles within an emergence angle of $\pm 2.7^\circ$. Hence, we neglect this effect.

The fact that the energy spectrum is not symmetric implies that the most probable energy loss is not equal to the average energy loss. Figure 8 (upper panel) shows the difference between the average energy loss $\langle \Delta E \rangle$ and the most probable energy loss $\Delta \hat{E}$ for nitrogen. We note that $\langle \Delta E \rangle$ is systematically larger than $\Delta \hat{E}$, as can be inferred from the residual energy distribution [i.e., Fig. 2(b)], which has a longer tail on the left side than on the right side. We also note that the difference $\langle \Delta E \rangle - \Delta \hat{E}$ decreases slowly with increasing energy and increases with foil thickness. Figure 8 (lower panel) shows the difference $\langle \Delta E \rangle - \Delta \hat{E}$ normalized to the

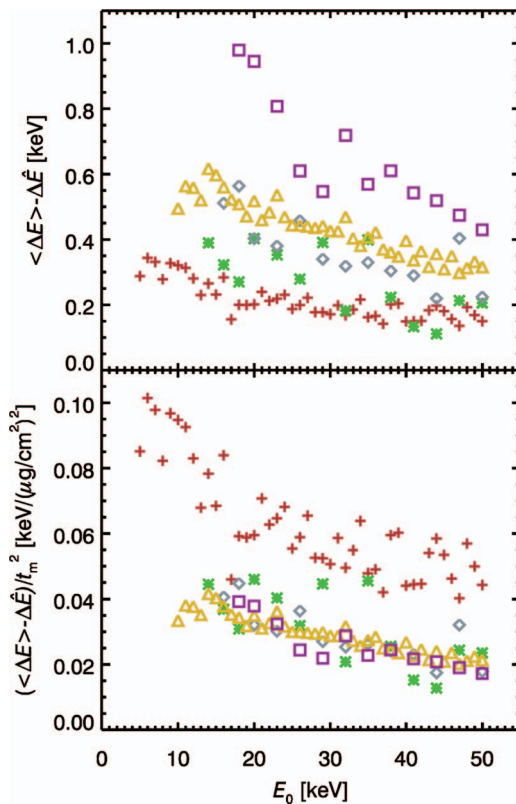


FIG. 8. (Color) Difference between average energy loss $\langle \Delta E \rangle$ and most probable energy loss $\Delta \hat{E}$ as a function of initial energy E_0 for nitrogen. $\langle \Delta E \rangle - \Delta \hat{E}$ is shown in the upper panel and the same normalized to the square of the foil thickness in the lower panel. Measured (t_m) and nominal (t_n) thicknesses are +, $t_m=1.84$, $t_n=0.5$; *, $t_m=2.96$, $t_n=1.0$; $\diamond$, $t_m=3.55$, $t_n=1.5$; $\triangle$, $t_m=3.85$, $t_n=2.0$; and $\square$, $t_m=5.00$, $t_n=3.0$ $\mu\text{g}/\text{cm}^2$.

square of the foil thickness. Except for the thinnest foil, the quantity $(\langle \Delta E \rangle - \Delta \hat{E})/t_m^2$ is the same for all foils, which suggests that $\langle \Delta E \rangle - \Delta \hat{E}$ is a function of the square of the foil thickness for nitrogen. Again, the larger fraction of the impurities on the thinnest foil might explain why the behavior is different from the other foils.

V. DISCUSSION

We showed that the average energy loss for H scales with foil thickness to the power 0.77. It is also implicit for other elements when looking at Fig. 5, except for argon in our energy range. However, in this energy range the nuclear stopping power of argon is larger than the electronic stopping power, which might explain the deviation. At low energies, we find that the energy loss of H seems to scale with foil thickness to the power 0.77, while the straggling of H seems to scale with the square root of the foil thickness. The difference between average energy loss and the most probable energy loss of nitrogen seems to scale with the square of the thickness. However, except for the energy loss straggling, these relationships do not apply to our thinnest foil. As mentioned above, the layers of impurities sticking on the surfaces of the foil might play an important role by changing its properties. The relative thickness of impurities is expected to increase with decreasing foil thickness. Another possible cause is that the thinnest foil might be too thin for the physical processes involved in the interaction between the primary particle and the foil to reach equilibrium.

Energy loss straggling is caused by a combination of many factors. Firstly, it is immanent to the statistical nature of the collisions. Secondly, foil thickness variations will naturally broaden the energy loss spectrum. Thirdly, the structure of the target, whether it is crystalline or amorphous, influences the energy loss spectrum. Fourthly, in a TOF measurement the electron TOF dispersion tends to broaden the energy loss spectrum. And finally, the beam energy dispersion also tends to broaden the energy loss spectrum.

In our analysis, we subtracted the electron TOF dispersion and neglected incident beam energy dispersion because it was very small compared to the energy straggling width. Both foil structure and thickness nonuniformity depend on the manufacturing process. According to the manufacturer, the carbon foils are amorphous (more accurately, nanocrystalline) carbon layers vacuum-arc-deposited onto a parting agent on a glass slide substrate. The manufacturer specifies the thickness nonuniformity being typically $\pm 5\%$ across a foil and the accuracy of the areal density measurement being $\pm 10\%$ or ± 0.5 $\mu\text{g}/\text{cm}^2$. Energy loss straggling measurements at various incident angles could be used to estimate the thickness nonuniformity of the foils.

One intriguing result is the dependence of $\langle \Delta E \rangle - \Delta \hat{E}$ on foil thickness. A higher value of $\langle \Delta E \rangle - \Delta \hat{E}$ is indicative of an asymmetric distribution. The large fluctuations in the energy loss are due to the small number of collisions involving large energy transfers. Deviation from a Gaussian distribution might arise from the following. The foil thickness is irregular. The foils are arc-deposited onto a parting agent on glass substrates at high vacuum. Due to the manufacturing pro-

cess, the carbon is deposited in nanosized aggregates, and, hence, thickness nonuniformity is expected. The foil thickness is therefore irregular based on the statistical deposition of carbon aggregates much larger than an individual atom. A thick foil should be mostly symmetric (e.g., Gaussian) because the statistical variation of the foil is decreasing relative to the mean thickness. For a thin foil, however, the statistical variation of thickness relative to the mean foil thickness is large, and one may expect a Gaussian energy loss distribution for the thicker parts of the foil and an additional, broader, and higher energy distribution of ions that transit thinner portions of the foil. Combined, this would look like a Gaussian distribution with a high energy tail. Evidence for this would be a more symmetric shape for thicker foils. However, we observe the opposite trend for nitrogen. Mertens²² measured $\langle E \rangle - \Delta \hat{E}$ for hydrogen at 100–300 keV in 10–130 nm carbon foils. He found that $\langle E \rangle - \Delta \hat{E}$ increases with energy and decreases with foil thickness, which is the opposite of what our measurements show. A possible explanation for this difference is that our measurements are done well below the stopping power maximum, whereas 100–300 keV hydrogen is above the maximum stopping power.

In conclusion, we measured the energy loss and straggling of H, He, C, N, O, Ne, and Ar ions passing through thin (nominal 0.5–0.3 $\mu\text{g}/\text{cm}^2$) carbon foils. Our main results are the following: (i) There seems to be a significant, roughly constant residue layer sticking to the surfaces of the carbon foils. We estimate the impurities' layer thickness to be about 1.6 $\mu\text{g}/\text{cm}^2$. However, we cannot say exactly how it is distributed in terms of which side. This might be important if sputtering is an issue. (ii) The energy loss seems roughly proportional to incident velocity. However, the energy loss measurements of H and He show a deviation from the velocity proportional electronic stopping power. Moreover, this proportionality does not work at low energy per mass, where the nuclear stopping power becomes more important than the electronic stopping power. (iii) The energy loss of H seems to vary with foil thickness to the power 0.77, except for our thinnest foil (nominal 0.5 $\mu\text{g}/\text{cm}^2$, measured 1.84 $\mu\text{g}/\text{cm}^2$). We derived simple empirical relations for the energy loss. (iv) The energy loss straggling ΔE_{FWHM} of H seems to vary with the square root of the thickness at low energies (below ~ 10 keV for H) and seems independent of foil thickness above ~ 20 keV (for H). The transition between these two regions is where $\Delta E/E_0$ is minimum. We derived simple empirical relations for the straggling for H and N. (v) The difference between the average energy loss and the most probable energy loss decreases with increasing energy. Except for our thinnest foil, it seems to vary as the square of the foil

thickness. (vi) The systematic differences between our thinnest foils and the thicker foils might arise from two sources. First, the thinnest foil has a larger amount of impurities on the surfaces in proportion to carbon thickness than the thicker foils. The impurities certainly have different physical properties than pure carbon. Second, for the very thin foils the statistical nature of the stopping process might be breaking down.

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